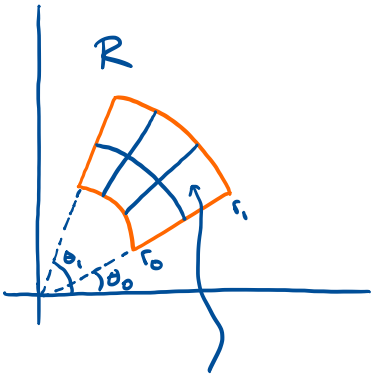
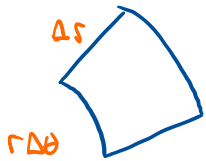
16.4 Double integrals in polar coordinates

Goal: Find an expression for dA in polar coordinates.



Suppose we have the polar region R divided into small sub-wedges of dimensions $r\Delta\theta$ by Δr. Then the area of the $(i,j)^{th}$ sub-wedge is

$$\Delta A \approx r\Delta r\Delta\theta.$$



$(i,j)^{th}$ sub-wedge

Letting the sub-wedge get smaller and smaller, in the limit we get

$$dA = r\,dr\,d\theta.$$

Thus if $z = f(r,\theta)$ is a function, we have

$$\int_R f\,dA = \int_{\theta_0}^{\theta_1}\int_{r_0}^{r_1} f(r,\theta)\,r\,dr\,d\theta.$$

Note: The limits of the integrals may be non-constant functions.

→ Do all problems.